CRSGD: Course Recommendation Framework Using Semantic AI and Generative AI Principles with Differential Control Metadata Scheme

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Abstract. The strategic course recommendation paradigm (CRSGD) is designed for semantics-oriented control, with a focus on the hybridization of AI and differential metadata selection. The approach leverages metadata capture to generate informative terms from datasets, using techniques such as LDA to model terms and categories. This process enables metadata generation from datasets, including e-books, to extract domain knowledge and support dataset analysis. Metadata classification within the Dspace enhances both dataset analysis and understanding of e-books. By using a deep learning RNN classifier and computing overlaps through strong semantic similarity measures, empirically determined thresholds are used to formalize these measures. The results are queried through controlled quantitative reasoning, which then achieves high performance metrics and establishes a best-in-class hybridized course recommendation strategy.

Keywords: Hybridization of AI, Deep learning RNN classifier, Semantic similarity measures.

1. Introduction

Many intelligent systems have developed rapidly because of the quick evolution in machine learning (ML) and artificial intelligence (AI) technologies, especially in the sector of education. More e-books, research papers, and multimedia content are available online than ever before, which has led to the increased demand for sophisticated course recommendation systems. Though they are good at making general recommendations, yet traditional recommendation systems frequently fail to fully utilize the complex domain knowledge that is present in all these kinds of resources. In addition, many Web 3.0 demands systems that combine various deep learning, semantic reasoning, and metadata production techniques to provide more personalized and intelligent suggestions to us. Thus, this research presents itself more like a hybridized approach of AI and a proper metadata selection model, which makes it able to serve as a foundation for a strategic course recommendation framework designed to address these issues.

The suggested system mainly combines the power of recurrent neural networks (RNNs) for deep learning-based classification with Latent Dirichlet Allocation (LDA) for topic modeling, in contrast to the conventional techniques, which are generally constrained by their inability to produce and categorize information dynamically. This system improves its e-book comprehension and dataset analysis by using cloud based large language models (LLMs) and a strong metadata generation approach, thus leading to more accurate course suggestions. The main component of this approach is the formalization of metadata overlaps that uses thresholds and semantic similarity measures, which enhances the classification accuracy and facilitates improved query processing inside the Dspace. This innovative contribution fills a significant void in current systems by providing a smoother and Web 3.0-compliant combination of knowledge augmentation and generative AI. Thus, this paper lays out a scalable and best-in-class approach for next-generation course recommendation systems using SPARQL querying and other agent-based reasoning processes.

Motivation: The lack of Web 3.0 compliant models that successfully combine generative AI with deep learning, knowledge augmentation, and progressive knowledge aggregation is the driving force behind the suggested architecture. This integration is very important when it comes to course recommendation systems, as it guarantees the coherence and compliance of Web 3.0. More than ever, there is a need for deep learning models that can create and categorize information properly. Current models frequently lack the capacity to construct semantic networks and use differential thresholds and deviation measures to compute information from various angles, which is why, this gap emphasizes the need for the suggested framework, which attempts to rectify these shortcomings and achieve the main goal of the study.

Contribution: The main contributions of this suggested system are the use of cloud-based LLMs for metadata production and the use of Latent Dirichlet Allocation for topic modeling along with caption generation from a dataset viewpoint. To main-

tain heterogeneity, this method generates metadata using a variety of technologies while including dense and enhanced e-book information. And to further identify the metadata, this system makes the use of a recurrent neural network (RNN) classifier, a reliable deep learning model with implicit feature selection. The other innovative contributions include formalizing a semantic network by applying the SimRank and Horns indices and estimating metadata overlaps with differential thresholds and step deviation measurements. Additionally, SPARQL allows for agent-based querying of the semantic network.

Organization: This paper follows a systematic structure, such that: In Section 2, an essential list of related works which helped to carry out the research work. Section 3 includes an overview of the proposed system design. In Section 4, the results are shown that are generated by the suggested mode. Section 5 ends the paper by going over the practical applicability of the model across a range of industries and their real-world implications.

2. Related Works

Development in artificial intelligence (AI) and machine learning techniques has led to developments in course recommendation systems. Burke [1] surveyed hybrid recommender systems and showed experimentally that combining collaborative and content-based techniques can offset the weaknesses of each, a strategy that later course recommenders adopted. Apaza et al. [2] applied Latent Dirichlet Allocation (LDA) to the syllabi of online courses and used the discovered topics as course descriptors in a content-based recommender, showing that topic models can organize large volumes of educational content. Van Deventer et al. [3] applied a large language model to a corpus of university course descriptions so that students can find courses through natural-language queries, illustrating how generative AI can add contextual understanding to course recommendation. Metadata has always been essential to such systems: Rizzo and Troncy [4] evaluated Linked Data entity extractors, including OpenCalais, on entity detection, typing and disambiguation, the capabilities that semantic metadata generation relies on. Sequential models have also been used for course selection: Pardos et al. [5] trained recurrent neural networks on enrollment histories to provide personalized course guidance at a large university and examined how scrutable those recommendations are. For graph-based similarity, Jeh and Widom [6] introduced SimRank, which considers two objects similar if they are related to similar objects, and Horn [7] proposed an index of overlap between two abundance distributions, now known as the Morisita-Horn index. Tarus et al. [8] reviewed ontology-based recommender systems for e-learning and showed how semantic web technologies improve the representation of learners and learning resources. Zhang et al. [9] proposed KGAN, which aggregates knowledge graph information for course recommendation in MOOCs and reported gains over state-of-the-art baselines. Klašnja-Milićević et al. [10] personalized e-learning with a hybrid recommendation strategy combined with learning style identification. Pardos and Jiang [11] used course descriptions, as auxil-

iary content beyond enrollment data, to surface relevant but unexpected courses in a university recommendation system. Finally, Zhang et al. [12] surveyed deep learning based recommender systems and their advantages in representation learning over traditional approaches.

3. Proposed Methodology

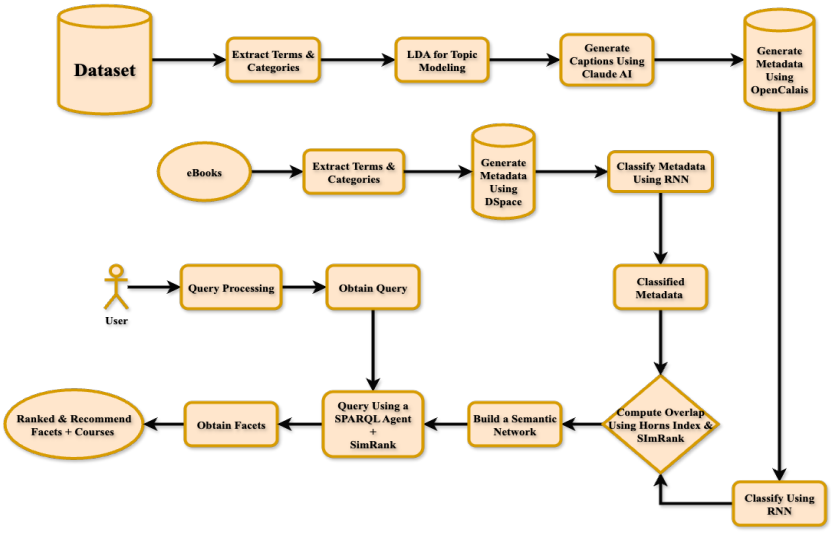


Fig. 1. System Architecture for the Proposed CRSGD Framework.

Fig. 1. illustrates the proposed system architecture for the strategy of the course recommendation framework, utilizing relative AI principles with metadata. This framework focuses on course recommendation. The dataset comprises courses, from which terms and categories are extracted. Since the categories are categorical in nature and not very informative independently, the Latent Dirichlet Allocation (LDA) model is applied. LDA was chosen as the topic modeling framework due to its robust and effective performance in extracting meaningful topics. Additionally, it integrates Web 3.0 instances directly into the proposed system's pipeline and framework. LDA plays a crucial role in topic discovery and modeling, ultimately reducing or lowering the dimensionality of the data. The LDA-discovered terms are extensively relevant but are limited in generation, which is handled using the Claude LLM. Once the captions are generated, Claude LLM yields enhanced captions, improving the density of auxiliary knowledge. However, additional information is still required, which is why OpenCalais is used to generate the LDA-discovered terms, Claude-generated captions, as well as terms from category extraction within the dataset. This process requires classification, which is achieved using an RNN classifier. RNN, being a strong deep learning classifier, is preferred due to its implicit feature selection capabilities.

Another important resource for knowledge generation is the eBooks relevant to the courses in the dataset. These eBooks undergo term and category extraction, where the contents of the eBooks are analyzed. The process can be enhanced by generating metadata using OpenCalais to introduce heterogeneity, which enables better utilization of the metadata. OpenCalais is favored for its robust ability to facilitate knowledge discovery through implicit feature selection techniques, and the inclusion of unknown objects from a different classifier supports metadata classification. Once classified, the metadata output from the pipeline and classifier undergoes computation to identify overlaps using Horns Index (with step deviation of 0.12) and SimRank (with a threshold of 0.75). Empirically, these thresholds are not compromised, ensuring that the metadata gains and the information from knowledge repositories and eBooks remain intact. This overlap helps in building a semantic network, creating a bridge between clusters formalized by Horns Index and SimRank. Wherever possible, single-link connections are established. The user query, which serves as the input, drives the system since this is a course recommendation framework. The query is subjected to preprocessing steps, including tokenization and lemmatization, before being processed by SPARQL agents through SimRank-controlled queries. SimRank, built using SPARQL queries, behaves according to a threshold of 0.75 to ensure accurate semantic and topic filtering. The results are then arranged in increasing order of SimRank measures, and upon clicking on the facets, the courses from the dataset are yielded based on term-category mapping, also organized in the same increasing order.

3.1. Latent Dirichlet Allocation (LDA)

A potent statistical model called Latent Dirichlet Allocation (LDA) is employed to identify abstract subjects in a set of documents. The model functions are based on the supposition that documents consist of multiple themes, with each topic having a distribution of words. To infer these hidden topic distributions from the observed documents is the main objective of LDA.

3.2. Claude AI

Claude AI is a collection of advanced language models developed by Anthropic with the goal of enhancing natural language production and understanding. Claude AI models use some of the state-of-the-art deep learning techniques to process and generate human-like language. These theories are based on various complex systems like transformers, which use attention mechanism to capture the complex relationships within text contents. The reputation of Claude AI for generating logical and contextually relevant language makes it useful for a very wide range of applications, including conversational bots, automated content production, and advanced text summarization processes. Although the architecture of Claude models is not public, they are built from many layers of neural networks with attention, which lets them understand and generate text in context. They are very efficient because they are trained on large volumes of data.

3.3. Horns Index

The Morisita-Horn index serves as a similarity measure for comparing traits or metadata distributions across datasets. Unlike conventional set-overlap metrics, it incorporates feature abundance rather than mere presence. Research indicates its effectiveness stems from quantifying distributional overlap through pairwise comparisons between feature weights of the used dataset. Scores range from 0 (no similarity) to 1 (identical distributions), with values typically represented through standardized formulas like Equation (1). Studies suggest its sensitivity to dominant features makes it robust against noise.

$$C_{MH} = \frac{2 \sum_{i=1}^{S} x_i y_i}{(d_x + d_y) XY} \tag{1}$$

where x_i is the weight (or frequency) of feature i in distribution X, y_i is the weight (or frequency) of feature i in distribution Y. $X = \sum_{i=1}^{S} x_i$ is the total weight of all features in X, $Y = \sum_{i=1}^{S} y_i$ is the total weight of all features in Y. $d_x = \frac{\sum_i x_i^2}{X^2}$. $d_y = \frac{\sum_i y_i^2}{Y^2}$ and S is the number of distinct features considered.

3.4. SimRank

The SimRank algorithm assesses node similarity in networks through structural link comparisons. Fundamentally, nodes are deemed analogous when their connected neighbors exhibit any comparable similarities. Research indicates applications in recommendation systems and social network analysis, particularly for identifying indirect relationships in graph-based datasets. This method mainly evaluates pairwise node relationships through recursive computation, where similarity scores depend on neighboring node likenesses. This recursive principle is formally captured in Equation 2. Notably, the approach avoids direct attribute comparisons, instead it relies on certain structural patterns to infer functional or relational parallels between various entities.

$$S(i, j) = \frac{C}{\left|N(i)\right| \left|N(j)\right|} \sum_{k \in N(i)} \sum_{l \in N(j)} S(k, l) \tag{2}$$

where $S(i, j)$ is the similarity between nodes i and j, $N(i)$ and $N(j)$ are sets of neighbors for nodes i and j respectively. C is a constant (usually between 0 and 1) that controls the decay factor of similarity. The sums are taken over all pairs of neighbors (k, l) where k is a neighbor of i and l is a neighbor of j.

4. Performance Evaluation and Results

The proposed approach utilizes a variety of datasets, including the Coursera-Course-Dataset [14], the E-Course Recommendation approach [13], and Udemy Courses and Reviews [15]. A Web 3.0 crawler is used to hybridize these datasets to collect comprehensive, subject-specific data for course suggestions. These datasets are improved and combined into a single, comprehensive dataset to ensure comprehensive coverage across academic areas. This augmentation process comprises merging data from many sources, standardizing format, and adding more metadata to the datasets to increase their utility. The resulting embedded dataset serves as the basis for complex calculations and metadata creation, and it increases the accuracy of topic modeling, metadata categorization, and semantic similarity measurements. The system overcomes the shortcomings of previous models by using this large dataset to provide extremely accurate course recommendations and semantic evaluations.

Evaluation protocol: All three baselines, GBCR [16], MCRS [17] and OBCR [18], were run on the same merged dataset as CRSGD, so that every model was evaluated on the same courses and queries. For each test query, human evaluators judged the recommended courses, and a recommendation was counted as relevant when it matched the topic or category of the query. Precision, recall, accuracy and F-measure were computed from these judgements, with the false discovery rate (FDR) reported as an auxiliary measure. Precision was also measured for the top 10 to 50 recommendations, as shown in Fig. 2.

Table 1. Comparison of Performance of the Proposed CRSGD with other approaches

Model	Average Precision %	Average Recall %	Average Accuracy %	Average F-Measure	FDR
GBCR [16]	89.44	90.15	89.80	89.79	0.11
MCRS [17]	90.04	92.08	91.05	91.05	0.10
OBCR [18]	93.19	94.08	93.63	93.63	0.07
Proposed CRSGD	95.85	97.08	96.47	96.46	0.05

The performance of the proposed Strategic Framework for Course Recommendation, which integrates Generative AI and Semantic Intelligence principles with differentially controlled metadata, is evaluated using various metrics: Average Precision, Average Recall, Average Accuracy, and Average F-Measure, along with the False Discovery Rate as an auxiliary standard metric. As shown in Table 1, CRSGD achieved the highest Average Precision of 95.85%, Average Recall of 97.08%, Average Accuracy of 96.47%, Average F-Measure of 96.46% with a corresponding False Discovery Rate of 0.05 which is the lowest among all. The selection of these metrics is crucial,

as they quantitatively assess the relevance and effectiveness of the recommendations generated by the framework.

Table 1 clearly indicates that CRSGD has outperformed all baseline models, establishing itself as the best-in-class solution. It achieved the highest Average Precision, Recall, Accuracy, and F-Measure, alongside the lowest False Discovery Rate (FDR). This outstanding result can be ascribed to CRSGD's knowledge-centric methodology, which uses differential principles in conjunction with a controlled metadata selection process to gradually improve course recommendations. The main reason why CRSGD performs better than any other baseline models is due to its extensive course dataset, which does offer a strong basis for training and assessment. Latent Dirichlet Allocation (LDA) modeling adds to the framework's rigor and hence guarantees a very strict examination of the structure and content of the course. Furthermore, the system also uses Claude to generate captions, which enables accurate and contextually appropriate course descriptions. Claude's hybridization of large language models (LLMs) makes it easier to comprehend language and context, and metadata generated in accordance with eBooks that are relevant to a certain subject creates a strong foundation for the suggestions. The integration of OpenCalais and DSpace, improves the model's overall heterogeneity by supplying rich metadata. In addition, CRSGD also uses computational methods like SimRank and Horns Index in conjunction with a usual RNN classifier to generate information. This combination efficiently manages querying via a SPARQL agent, producing the best-in-class facets and guaranteeing that the courses that are suggested are not only pertinent but also in line with the requirements and preferences of the users.

The GBCR [16] model, on the other hand, falls short of performance expectations when compared to the suggested framework. This is mainly because it is a personalized course recommendation system that is goal-based. It uses previously recorded and preserved data to try to capture students' unique interests. Even though the model's structured approach results in some degree of differential overlap and better personalized recommendations, it still doesn't address important issues like serendipity and cold beginnings. The GBCR [16] model lacks robust inferential learning schemes, and is overly dataset dependent, which restricts its adaptability. As a result, it performs substantially worse than the suggested structure. Its effective personalization contributes to a reasonably high relevance rate; however, this benefit is offset by its many drawbacks. The model needs the addition of auxiliary knowledge and inferential learning strategies, such as machine learning models and semantic similarity metrics, to get beyond these constraints.

Because the MCRS [17] model is primarily concerned with course recommendation for MOOCs and depends on distributed computing, it is unable to adequately address the proposed framework. Although course selection is based on association rule mining—more precisely, an enhanced A-Priori algorithm—the mining of these rules is carried out via computationally reliable frameworks such as Spark and SCOOP. Nevertheless, even when a certain degree of computational stability is at-

tained, the computations get more complex, and the model is not enhanced with more information from outside sources. Its ability to learn and reason inferentially is restricted by this absence, which is necessary for it to adjust to the changing terrain of course recommendations. While the enhanced A-Priori method derived from association rule mining facilitates course recommendation decision-making, it is still limited to a single dataset. This makes the necessity of augmenting auxiliary knowledge evident. The inferential learning reasoning powers of the MCRS [17] model are insufficient, leading to subpar performance.

In a similar vein but being an ontology-based personalized post-recommendation framework, the OBCR [18] model likewise performs poorly. It mainly solves cold start and diversity difficulties, while it also adds augmented auxiliary knowledge and personalization through user historical data. The model combines content-based filtering and collaborative filtering, yet this hybrid strategy is insufficient. The fundamental challenges of collaborative filtering are not sufficiently addressed by the combination of user ratings and content-based filtering, which depends on the actual similarity of the content. Moreover, flexibility is limited by the fact that although OBCR [18] uses dynamic ontology mapping, the ontologies that are employed are preset. Although some supplementary knowledge is used, its usefulness is limited when it is linked only by keyword similarity. Despite ontologies adding a certain density of auxiliary information, the dataset and a thorough examination of the course material point to the urgent need for development. The present model is severely limited because it does not have the high-density auxiliary knowledge needed for thorough course recommendations.

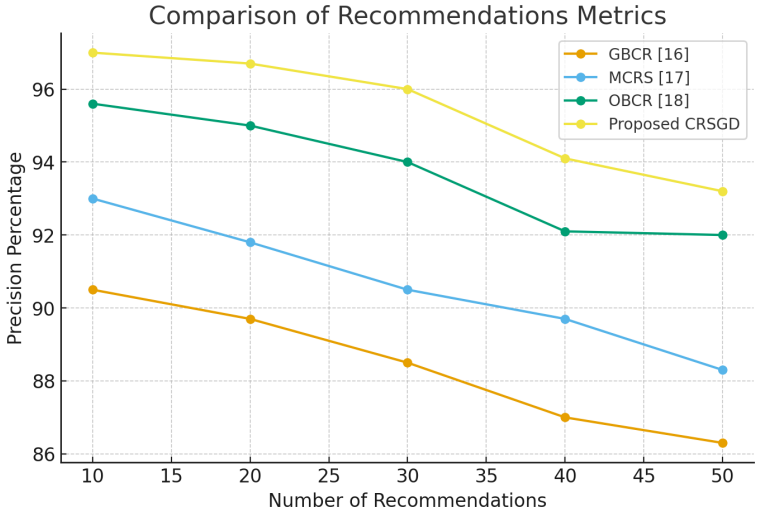


Fig. 2. Precision Percentage vs. Number of Recommendations

Fig. 2 illustrates the distribution curve of the number of recommendations generated by the proposed CRSGD model in comparison to baseline models GBCR [16], MCRS [17], and OBCR [18]. The graph clearly demonstrates that CRSGD occupies the highest portion of the hierarchy, indicating its superior performance in generating recommendations. Conversely, GBCR occupies the lowest position in the hierarchy, reflecting its limitations in providing effective course recommendations. The primary reasons for CRSGD's leading position include its comprehensive dataset, the integration of LDA for stringent content analysis, the use of Claude for accurate caption generation, and the hybridization with large language models that enhance contextual understanding. Additionally, the framework's ability to generate relevant metadata through advanced techniques contributes significantly to its effectiveness. In the middle tier, OBCR [18] ranks second, showcasing moderate performance; however, it still falls short of the capabilities demonstrated by CRSGD. MCRS [17], positioned just below OBCR [18], struggles to maintain competitiveness due to its inherent complexities and lack of adequate auxiliary knowledge augmentation.

The data visualization shows the notable advantages of the CRSGD model while exposing certain persistent challenges in existing approaches like GBCR [16], MCRS [17] and OBCR [18] regarding personalization accuracy. Notably, comparative analysis reveals those consistent performance gaps that align with recent studies on collaborative filtering limitations. Therefore, these findings do contribute to the ongoing discussions about optimizing recommendation systems through adaptive feature weighting.

5. Conclusion

The proposed architecture combines various deep learning and generative AI techniques in a very novel way, thus, providing a very robust solution for the Web 3.0 course recommendation systems. By merging cloud-based metadata creation, Latent Dirichlet Allocation for topic modeling, and recurrent neural networks for classification, it succeeds in addressing the drawbacks of the previous methods. Henceforth, this method ensures scalable metadata classification and knowledge aggregation, constructs certain semantic networks, and uses differential thresholds and SPARQL-based querying. Some of the useful applications include optimizing corporate training programs, streamlining course curation for academic institutions, and enhancing learning environments with customized suggestions. This adaptable, cutting-edge technology meets all the necessary requirements of Web 3.0 learning environments.

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